

EXHIBIT 2

Affidavit of Dr. Kirsten Smith

IN THE CIRCUIT COURT OF JACKSON COUNTY, MISSOURI
AT KANSAS CITY

STATE OF MISSOURI, et al.

Plaintiffs,

v.

Case No. 2616-CV11773

SHAMAN BOTANICALS, LLC, et al.

Defendants.

AFFIDAVIT OF DR. KIRSTEN SMITH, Ph.D., L.M.S.W

I, Dr. Kirsten Smith, being duly sworn, declare under oath as follows:

1. I am an adult individual competent to make this affidavit and have personal knowledge of the facts stated herein.
2. I am a trained clinician and researcher with training and experience in substance use disorder ("SUD") assessment and treatment, behavioral pharmacology research, and the clinical and behavioral study of kratom (*Mitragyna speciosa*) and kratom-derived products, including 7-hydroxymitragynine ("7-OH").
3. I earned my Bachelor's degree in Social Work from the University of Kentucky in 2014. I earned a clinical Master's degree in Social Work ("MSW") from the University of Kentucky in 2015.
4. After earning my MSW, I assisted on research and data projects at the University of Kentucky's Center on Drug and Alcohol Research ("CDAR"), including evaluation of outcomes for Kentucky's corrections-based drug treatment program.
5. I first became aware of the botanical *Mitragyna speciosa* ("kratom") in 2015.
6. I completed a joint Ph.D. program in social work at the University of Louisville and the University of Kentucky in 2019. My doctoral work focused on social determinants of health related to drug use and criminal justice involvement.
7. During my Ph.D. training, I undertook one of the earliest U.S.-based investigations of kratom use among adults with drug use histories enrolled in treatment, resulting in an original research publication (Smith & Lawson, 2017).

8. Following completion of my Ph.D., I began a four-year postdoctoral training fellowship at the National Institute on Drug Abuse Intramural Research Program ("NIDA") in Baltimore, Maryland, which I completed in July 2023.

9. While at NIDA, I collaborated with researchers interested in understanding kratom use and expanded my training in kratom-related pharmacology.

10. In 2021, I was awarded a NIDA research grant to investigate kratom.

11. I have provided oral and written testimony and presentations on kratom to state legislative and regulatory bodies throughout the United States.

12. I have approximately 90 peer-reviewed papers published or in press, with additional manuscripts under review or in active preparation. The majority of my peer-reviewed work pertains directly to kratom.

13. I have reviewed Plaintiffs' Petition and accompanying affidavits, including the Affidavit of Dr. Igor Koturbash submitted in support of Plaintiffs' Motion for Temporary Restraining Order and Preliminary Injunction.

14. In my opinion, it is not scientifically accurate to treat the profile of kratom as interchangeable with the profile of 7-OH.

15. Kratom is a botanical and 7-OH is a kratom alkaloid and metabolite.

16. Neither kratom nor 7-OH is an "opiate."

17. Neither kratom nor 7-OH has been demonstrated to function like an opiate in human consumers.

18. I am not aware of any scientific evidence demonstrating that 7-OH is more potent than morphine in humans.

19. I am not aware of any evidence demonstrating that respiratory depression has been casually attributable to kratom or 7-OH.

20. In my opinion, the animal literature pertaining to kratom or 7-OH to date does not, by itself, establish serious safety concerns under typical real-world oral use conditions.

21. From a clinical and scientific perspective, terms such as "addiction," "physical dependence," "tolerance," and "withdrawal" are distinct and are not interchangeable.

22. Claims that kratom use "leads to tolerance" or that kratom products are "chemically addictive" are overbroad as categorical statements. In my experience and based on the scientific literature, some individuals who use kratom do not develop tolerance, physical dependence, or addiction, and risk varies based on factors including dose, frequency, duration of use, product type, and individual vulnerability.

23. The phrase "chemically addictive" is vague and is not a standard term of art used in my field to characterize a substance's abuse liability or SUD risk.

24. Many products or substances that are experienced as rewarding by some individuals—including products used by the average consumer (i.e. caffeine)—can be associated with some risk of problematic use or SUD in a subset of users or may be used in a normative manner without the development of an SUD.

25. Kratom and 7-OH have been used in the United States for years, yet based on my review of the available evidence, there is limited evidence of widespread development of severe SUD or addiction attributable to kratom or 7-OH.

26. In my opinion, the existence of some abuse potential or SUD risk does not, by itself, establish that a product is inherently dangerous or that its use poses an urgent public health crisis.

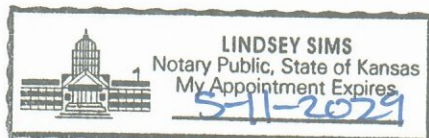
27. I do not agree kratom or 7-OH use contributes to homelessness, vagrancy, or crime. In my opinion, those assertions are not established by the clinical and scientific evidence I have reviewed and are not supported by my experience studying kratom use patterns and motivations among U.S. consumers.

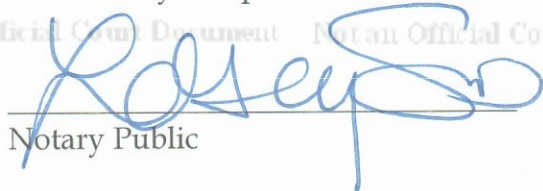
28. As evidenced by the overall lack of data, it is my opinion that kratom and/or 7-OH do not pose an unreasonable risk of illness or injury, particularly relative to many other available substances.



Dr. Kirsten Smith, MSW, Ph.D.

Subscribed and sworn before me this 20th day of April 2026.




Notary Public